Assignment 1 - Git & GitHub Fundamentals

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GitHub account handle: USERNAME_PENDING. Display name supplied: Hafsa mhf.

> DRAFT FOR FINALIZATION: Local Git work is complete. GitHub publication, the online README edit, fetch/pull evidence and URL submission remain pending. Review and update this report before submission.

1. Repository Setup

The provided STARTER files were copied into a new repository and initialized with git init -b main. The final .gitignore contains local.env, *.log, and !important.log. The first two rules ignore the local configuration and ordinary logs; the exception keeps important.log available to Git.

A simple issue was demonstrated and solved: the broad *.log pattern initially ignored important.log. Adding !important.log corrected that result. Before/after command captures are stored in evidence/01_ignore_before and evidence/02_ignore_after.

Final ignore rules: local.env; *.log; !important.log.

2. README & Commit History

README.md contains my name, student ID, course/section, the supplied display name, a project description, a technologies list, a Bash code block, a documentation link, and the provided assets/project-banner.svg image. USERNAME_PENDING marks the exact GitHub handle that must be confirmed.

Five content commits record setup, the overview improvement, CSS styling, the JavaScript status message, and the About feature. An explicit merge commit integrates the feature. The PDF and evidence are added in an additional commit after the workflow captures.

Commit	Meaningful change
86c89fe	Initialize starter project with README and ignore rules
92e9dfa	Clarify the project overview
d2ad1f8	Improve section spacing and borders
2e57c24	Update the project status message
813ba1c	Add the About feature on its own branch
a46ee83	Merge the About feature into main

3. Selective Staging

After the initial commit, index.html and style.css were modified together. Only index.html was staged with git add index.html and committed first. The remaining CSS modification was confirmed with git status and git diff before a separate CSS commit.

The evidence includes a full unstaged diff and staged diff. In the status excerpt, M in the left column for index.html means staged; M in the right column for style.css means modified but unstaged.

```
Selective staging: only index.html is staged; the CSS change remains unstaged. Both diff types are
 shown.
Actual output captured in the AI-assisted execution environment.
UTC: 2026-10-12T12:02:58+00:00

$ git status --short
M  index.html
 M style.css
```

Actual Git output, rendered as an excerpt. Full staged/unstaged diffs are in evidence/04_selective_staging.txt. These captures were generated in the assistant environment.

```
After the HTML-only commit, style.css remains modified and uncommitted.
Actual output captured in the AI-assisted execution environment.
UTC: 2026-10-12T12:02:58+00:00

$ git status --short
 M style.css

$ git diff -- style.css
diff --git a/style.css b/style.css
index b93e076..fae9c24 100644
--- a/style.css
+++ b/style.css
@@ -19,3 +19,11 @@ section {
   padding: 20px;
   border: 1px solid #e3e3e3;
 }
+
+/* Make individual sections easier to distinguish. */
+main section {
+  padding: 1rem;
+  margin-bottom: 1rem;
+  border: 1px solid #d5dce5;
+  border-radius: 6px;
+}
```

After committing index.html, the CSS change remains modified. It was subsequently committed separately.

4. Branch & Merge

The feature-about branch was created with git switch -c feature-about. An About section was added to index.html and committed on that branch. After switching back to main, the feature was merged using git merge --no-ff feature-about. The explicit merge commit makes the branch integration visible in the graph.

The About section was verified to be absent from main before merging and present afterward. The feature branch is retained for inspection.

```
The About feature has been merged into main with an explicit merge commit.
Actual output captured in the AI-assisted execution environment.
UTC: 2026-10-12T12:02:58+00:00

$ git branch --list
  feature-about
* main

$ git log --graph --oneline --decorate --all
*   a46ee83 (HEAD -> main) Merge the About feature into main
|\
| * 813ba1c (feature-about) Add the About feature on its own branch
|/
* 2e57c24 Update the project status message
* d2ad1f8 Improve section spacing and borders
* 92e9dfa Clarify the project overview
* 86c89fe Initialize starter project with README and ignore rules
```

The graph captures the actual history before the report/evidence commit.

Ignore-rule verification

```
Troubleshooting resolution: the exception keeps important.log visible while normal logs and local.env remain
ignored.
Actual output captured in the AI-assisted execution environment.
UTC: 2026-10-12T12:02:58+00:00

$ git status --short --ignored
?? .gitignore
?? app.js
?? assets/
?? important.log
?? index.html
?? notes.txt
?? style.css
!! app.log
!! debug.log
!! local.env

$ git check-ignore -v local.env app.log debug.log
.gitignore:1:local.env  local.env
.gitignore:2:*.log      app.log
.gitignore:2:*.log      debug.log
```

Normal log files and local.env are ignored; important.log remains available and was tracked in the initial commit.

5. Remote Changes

A commit saves staged changes in the local repository; push publishes local commits to a remote. Fetch downloads remote information and updates remote-tracking references without changing working files. Pull fetches and integrates remote changes into the current branch.

PENDING: create the correctly named empty GitHub repository, publish main, edit README directly on GitHub, fetch and capture local main behind origin/main before pulling, then pull and verify the edited sentence. These operations have not been performed. The provided Finish_GitHub.sh guide records the actual evidence when they are completed.

6. Fair Use of AI

AI assistance was used to interpret the assignment, prepare the README and source edits, execute the local Git workflow, and draft this report after my laptop became unresponsive. My prompts asked for a step-by-step explanation and then an assignment package. The generated work was checked by the assistant using actual Git status, diff, log, ignore-rule, branch and merge outputs. Those checks were performed in the assistant's execution environment, not on my laptop. Before submission I must review the commands and verify the actual GitHub operations. The commits use my supplied name and a temporary local email (git-assignment@localhost); the exact GitHub handle and my account email still need confirmation.

7. Sources & Reflection

Sources: Assignment 1 and Report Template - group A.pdf; Git presentation Session 1 - group A.pdf; the supplied Assignment1_Starter archive; GitHub documentation about valid account handles. The report and evidence should be reviewed before submission.

Draft reflection - personalize before submitting: The difference between fetch and pull is a concept I need to confirm: fetch updates remote knowledge, while pull integrates changes. If Git behaves unexpectedly, I would first inspect git status, the active branch, and git remote -v.

Evidence format: command outputs were captured from real executions, saved verbatim as text, and rendered into readable images. No GitHub operations or student-laptop screenshots are represented as completed when they were not.